

BILLUND AP, Denmark

WMO: 061040

Lat: 55.738N

Lon: 9.168E

Elev: 75

StdP: 100.42

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-7.1	-11.4	1.4	-7.4	-9.5	1.7	-5.6	14.2	6.1	12.8	6.1	1.5	120	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.9	18.3	24.9	17.5	22.9	16.6	19.5	24.1	18.5	22.8	17.6	21.5	4.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.1	21.1	17.0	12.2	20.1	16.0	11.5	18.9	55.9	24.0	52.5	22.8	49.6	21.5	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.7	8.6	DB	-13.5	29.4	3.1	1.6	-15.7	30.5	-17.5	31.5	-19.2	32.4	-21.4	33.6	(4)
(5)			WB	-13.7	21.3	3.0	1.1	-15.9	22.0	-17.6	22.7	-19.3	23.3	-21.5	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	0.9	1.1	3.1	7.3	11.1	13.9	16.5	16.4	13.2	9.1	4.9	2.2	(6)
(7)		DBStd	6.55	3.94	3.84	3.48	3.46	3.32	2.85	2.94	2.86	2.69	3.20	3.45	4.19	(7)
(8)		HDD10.0	1319	281	248	214	94	26	2	0	0	4	54	154	242	(8)
(9)		HDD18.3	3680	539	482	472	330	224	137	76	76	156	286	402	500	(9)
(10)		CDD10.0	719	0	0	0	14	61	119	200	198	99	26	2	0	(10)
(11)		CDD18.3	39	0	0	0	0	1	4	18	15	1	0	0	0	(11)
(12)		CDH23.3	333	0	0	0	1	10	44	158	114	6	0	0	0	(12)
(13)		CDH26.7	49	0	0	0	0	1	5	26	18	0	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.6	4.7	4.6	4.1	4.0	4.1	3.6	3.5	3.7	3.9	3.9	4.1	(14)
(15)	Precipitation	PrecAvg	925	86	64	55	45	53	70	76	95	91	102	90	97	(15)
(16)		PrecMax	1136	164	143	124	94	103	147	134	185	184	251	179	196	(16)
(17)		PrecMin	617	4	21	10	6	6	1	18	29	17	37	26	26	(17)
(18)		PrecStd	134	40	31	29	21	25	35	33	44	44	42	39	48	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.8	14.1	21.8	24.8	27.2	29.2	28.7	24.1	17.8	13.8	10.9	(19)
(20)			MCWB	8.3	8.2	8.4	13.7	16.5	18.3	19.3	18.9	17.2	14.8	12.6	10.2	(20)
(21)		2%	DB	8.0	8.1	11.2	18.2	22.1	24.4	27.1	26.4	21.2	15.9	12.0	9.7	(21)
(22)			MCWB	7.4	7.2	7.4	11.5	14.7	17.0	18.4	18.3	16.0	13.7	11.0	9.1	(22)
(23)		5%	DB	7.1	7.2	9.2	15.8	20.0	22.0	25.1	24.1	19.2	14.7	10.9	8.7	(23)
(24)			MCWB	6.3	6.4	6.7	10.3	13.6	15.8	17.8	17.5	15.0	13.2	9.9	8.1	(24)
(25)		10%	DB	6.1	6.2	7.8	13.2	17.9	19.9	22.9	22.0	17.8	13.7	9.8	7.7	(25)
(26)			MCWB	5.4	5.4	6.2	9.1	12.5	14.9	16.9	16.7	14.5	12.4	9.0	7.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	8.4	9.7	14.3	17.5	19.7	21.0	20.7	18.4	15.6	12.8	10.3	(27)
(28)			MCD B	9.0	8.9	11.7	20.2	23.1	24.4	26.1	25.1	22.0	17.1	13.6	10.7	(28)
(29)		2%	WB	7.4	7.4	8.2	12.4	15.9	18.0	19.7	19.4	16.8	14.4	11.2	9.2	(29)
(30)			MCD B	7.8	7.9	10.0	17.0	20.2	22.7	24.7	23.5	19.7	15.3	11.9	9.6	(30)
(31)		5%	WB	6.4	6.5	7.2	10.9	14.4	16.6	18.6	18.4	15.9	13.4	10.1	8.1	(31)
(32)			MCD B	6.8	7.0	8.5	14.6	18.4	20.7	23.0	22.5	18.2	14.4	10.7	8.4	(32)
(33)		10%	WB	5.4	5.5	6.3	9.8	13.1	15.4	17.5	17.5	15.1	12.5	8.9	7.1	(33)
(34)			MCD B	5.9	6.0	7.5	12.6	17.1	18.8	21.6	21.1	17.1	13.5	9.5	7.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	4.8	6.6	8.8	9.5	8.8	9.3	9.1	7.7	6.2	4.9	4.4	(35)
(36)			MCDBR	4.2	4.8	8.9	12.4	13.1	12.8	12.9	12.2	9.8	6.8	4.7	3.9	(36)
(37)			MCWBR	4.3	4.1	5.8	6.7	7.0	6.7	5.8	5.5	5.5	5.1	4.4	4.0	(37)
(38)		5% WB	MCDBR	4.1	4.4	7.3	10.8	11.2	11.0	11.0	10.1	8.0	5.9	4.6	4.0	(38)
(39)			MCWBR	4.2	4.1	5.4	6.4	6.7	6.5	5.7	5.3	5.1	4.9	4.6	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.337	0.360	0.394	0.386	0.381	0.396	0.397	0.374	0.353	0.331	0.304	(40)	
(41)		taud	2.239	2.282	2.283	2.241	2.315	2.352	2.327	2.336	2.385	2.381	2.298	2.232	(41)	
(42)		Ebn at Noon	578	721	798	822	855	863	840	813	779	698	560	499	(42)	
(43)		Edh at Noon	57	79	99	120	118	115	116	108	90	71	54	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.48	1.09	2.42	4.03	5.27	5.60	5.20	4.15	2.87	1.48	0.60	0.35	(44)	
(45)		RadStd	0.08	0.15	0.22	0.48	0.51	0.44	0.56	0.31	0.30	0.18	0.09	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	-117	-163	N/A	N/A

Nomenclature: See separate page